Orso Forghieri

PhD in Reinforcement Learning

8 rue Maison Dieu
75014 Paris, France
M *+33 6 61 32 95 38*
E *orso.forghieri@gmail.com*
https://orsof.github.io/

Research Profile

Researcher in Reinforcement Learning and Applied Mathematics, working at the intersection of **Reinforcement Learning**, **Dynamic Programming**, **Markov Decision Processes**, **state abstraction**, and **hierarchical methods** for scalable decision-making in high-dimensional environments.

Core topics: State abstraction; hierarchical reinforcement learning; approximate dynamic programming; planning; Markov aggregation/disaggregation; scalable MDP solving; policy iteration; value iteration; Q-value iteration; stochastic optimization.

Applications: Large-scale planning, resource allocation, network optimization, edge computing, service placement, server energy and performance optimization, market forecasting, and railway-delay propagation modelling.

Education

Sep. 2022 – Dec. 2025 **Ph.D. in Applied Mathematics / Reinforcement Learning**, École polytechnique, Institut Polytechnique de Paris, Palaiseau, France.
Thesis: *Hierarchical Reinforcement Learning for Large Scale Problems*.
Supervision: Erwan Le Pennec, Hind Castel-Taleb, Emmanuel Hyon.
Laboratory: Centre de Mathématiques Appliquées, École polytechnique.
Research: hierarchical reinforcement learning, state abstraction, progressive disaggregation, scalable dynamic programming, large-scale MDPs.

Sep. 2021 – Aug. 2022 **MSc MVA – Mathematics, Vision, Learning**, École Normale Supérieure Paris-Saclay, Gif-sur-Yvette, France.
Thesis: *Hierarchical Reinforcement Learning for Optimizing Large Systems*, supervised by Erwan Le Pennec, CMAP.
Selected coursework: time series analysis, deep learning, reinforcement learning, object recognition.

Sep. 2018 – Aug. 2021 **Diplôme d'ingénieur**, GPA: 3.7, École polytechnique, Palaiseau, France.
Thesis: *Mobility in Epidemiological SEIR Model and Application to the COVID-19 Propagation from Italy to France*, supervised by Stéphane Gaubert, CMAP.
Project: *An Overview of Market Forecast with Machine Learning Techniques*, supervised by Guillaume Hollard, CREST, in collaboration with Lusis Solutions.
Project: *Solving Open Shortest Path First Problems with Evolutionary Methods*, supervised by Eric Gourdin, Orange Gardens.
Selected coursework: statistical learning, uncertainty quantification, risk analysis, functional analysis, algorithms, machine learning, deep learning.

Sep. 2015 – Sep. 2018 **Classe Préparatoire aux Grandes Écoles MPSI–MP**, Lycée Louis-le-Grand, Paris, France.
TIPE: *Location of a Photographer using Projective Geometry Methods*, supervised by Jean-Pierre Roundneff.
Admission to École polytechnique.

Research and Work Experience

Feb. 2026 — Today **Research Collaboration, École Polytechnique**
- Designed and implemented adaptive learning algorithms for sequential decision-making, including AggQL, a model-free RL method that learns and refines its internal state representation directly from interaction using confidence-controlled TD-residual tests.
- Developed Confidence-Adaptive Expert Disaggregation (CAED), a parameter-efficient LLM adaptation method that dynamically creates LoRA specialists using confidence-controlled expert-creation criteria based on statistically significant gradient disagreement; derived an optimization bound quantifying the benefit of expert specialization.
- Developed mdpforge, a modular Python framework for reproducible benchmarking of finite MDP solvers, with matched accuracy and independent error checks.
- Built reproducible Python/Gym environments and benchmarking pipelines for evaluating exact, approximate, and learning-based decision-making algorithms.

Oct. 2025 – Feb. 2026 **Quantitative Researcher, Qube Research & Technologies, Paris, France.**
- Developed sequential and explainable decision-making methods for systematic equity trading.
- Worked on interpretable modelling of trading decisions for equity-market applications.
- Applied reinforcement-learning and statistical-learning methods to financial decision processes.

2022 – 2026 **PhD Researcher in Reinforcement Learning, CMAP, École polytechnique, Palaiseau, France.**
- Developed abstraction-refinement and progressive state-space disaggregation methods for large Markov Decision Processes.
- Studied the links between approximate dynamic programming, projected Bellman operators, and abstract MDPs.
- Integrated disaggregation procedures into approximate value iteration, Q-value iteration, and policy iteration.
- Benchmarked methods on random MDPs, Four Rooms, Mountain Car, Sutton's racetrack, tandem queues, and hydro-valley management models.

2021 – 2025 **Research Collaboration in Reinforcement Learning and Combinatorial Optimization, Orange Gardens / CMAP.**
- **Edge service placement and routing:** modelled latency-constrained orchestration under local-information, network-flow, and compute-capacity constraints for industrial 6G use cases including Smart Warehouse and Augmented Reality.
- Designed a lightweight hybrid orchestration approach combining rule-based mechanisms with RL-based service placement, enabling deployment on operational instances with local observations.
- Achieved placement and routing decisions on the order of 1 s and up to approximately 50% expected latency reduction in simulation compared with the reference strategy.
- **VM management and energy optimization:** developed a scalable method to extract simplified and interpretable dynamics from servers with time-varying workloads and on the order of one hundred VMs.
- Derived interpretable VM-management policies yielding approximately 30% expected energy savings in simulation.
- Publications and communications: *Service Placement under Affine Delay Constraint for Computational Resources Management*, ROADEF 2022; *Faster Latency Constrained Service Placement in Edge Computing with Deep Reinforcement Learning*, IFIP Networking 2025.

Mar. 2022 **Research Intern in Hierarchical Reinforcement Learning, Télécom SudParis.**
- Explored server energy and performance optimization under uncertainty using hierarchical reinforcement learning.
- Work conducted in collaboration with Energy4Climate.

Jun. 2020 – Sep. 2020 **Machine Learning Intern, SNCF Réseau – DGEX Solutions**
- Developed Python machine-learning models to predict train delay propagation and estimate delayed train passage times for passenger information and traffic regulation.
- Engineered features from historical train circulation data and compared predictive models against existing delay-forecasting baselines at network scale.
- Shared code through GitLab and presented technical results to DGEX Solutions stakeholders, focusing on model relevance for network resilience and robustness.

Sep. 2019 – May 2020 **Research Project in Market Forecasting, Lusis Trading Solutions**

- Developed machine-learning models for financial market forecasting on CAC40, UK100, and commodity time series, using CNN and LSTM architectures.
- Integrated financial news signals through sentiment analysis and Word2Vec-based NLP features from Bloomberg and Reuters headlines.
- Analysed model limitations including class imbalance, delayed predictions, exploding/vanishing gradients, and loss-function design for directional forecasting.

Publications

2025 Orso Forghieri, ***Hierarchical Reinforcement Learning for Large Scale Problems***, PhD thesis, Institut Polytechnique de Paris, Palaiseau, France.

2025 Orso Forghieri, Yannick Carlinet, Emmanuel Hyon, Erwan Le Pennec and Nancy Perrot, **Faster Latency Constrained Service Placement in Edge Computing with Deep Reinforcement Learning**, *IFIP Networking*, 2025.

2024 Orso Forghieri, Hind Castel, Emmanuel Hyon and Erwan Le Pennec, **State Abstraction Discovery from Progressive Disaggregation Methods**, *17th European Workshop on Reinforcement Learning (EWRL)*, Toulouse, France, 2024.

2024 Orso Forghieri, Hind Castel, Emmanuel Hyon and Erwan Le Pennec, **Progressive State Space Disaggregation for Infinite Horizon Dynamic Programming**, *International Conference on Automated Planning and Scheduling (ICAPS)*, 2024, pp. 221–229.

Manuscripts and Preprints

- Orso Forghieri, Emmanuel Hyon, Hind Castel and Erwan Le Pennec, **Confidence-Adaptive State Abstraction for Model-Free Q-Learning**, submitted to AISTATS 2027.
- Orso Forghieri, Hind Castel, Emmanuel Hyon and Erwan Le Pennec, **Value-Aware State Disaggregation for Continuous-State MDPs**, submitted to AISTATS 2027.
- Orso Forghieri, Hind Castel, Emmanuel Hyon and Erwan Le Pennec, **Approximate Dynamic Programming and State Abstraction Discovery**, manuscript under peer review.
- Orso Forghieri, Hind Castel, Emmanuel Hyon and Erwan Le Pennec, **Disaggregation Techniques for Faster Markov Decision Process Solving and State Abstraction Discovery**, experimental code repository associated with a submitted manuscript.

Conference Communications and Presentations

Jun. 2025 Talk at the SIMPAS Team Retreat, Rouen, France.

Nov. 2024 Talk at PGMO Days, EDF Labs, Palaiseau, France: *State Abstraction Discovery for Infinite Horizon Dynamic Programming*.

Sep. 2024 Talk at the Argo Seminar, Inria Paris, France.

Apr. 2024 PhD seminar, Centre de Mathématiques Appliquées, École polytechnique, Palaiseau, France.

Mar. 2024 Two sessions at ROADEF 2024, Amiens, France:

- *Progressive State Space Aggregation/Disaggregation for Dynamic Programming*.
- *Determining Efficient Methods for Solving Markov Decision Processes*, with Emmanuel Hyon.

Nov. 2023 Presentation in the Young Researchers session at the Grenoble Workshop on Restless Bandits, Index Policies and Applications in Reinforcement Learning.

Jan. 2023 Talk at the SIMPAS team seminar, CMAP, École polytechnique.

Feb. 2022 Session at ROADEF 2022: *Service Placement under Affine Delay Constraint for Computational Resources Management*.

Teaching Experience

Sep. 2024 Teaching Assistant in Statistics, Engineering degree, École polytechnique, course by Eric Moulines.

Sep. 2024 Teaching Assistant in Probabilities, MSc, École polytechnique, course by Tabea Rebafka.

Mar. 2024 Teaching Assistant in Rare Events Numerical Simulation, Engineering degree, École polytechnique, with Gersende Fort, Tony Lelièvre and Arnaud Guyader.

Sep. 2023 Teaching Assistant in Statistics, Engineering degree, École polytechnique, course by Eric Moulines.

Sep. 2023 Teaching Assistant in Probabilities, MSc, École polytechnique, course by Tabea Rebafka.

Feb. 2023 Teaching Assistant in Operational Research, BSc, Université Paris Nanterre, course by Claire Hanen.

Academic Service, Outreach and Professional Activities

2024 – 2025 PhD Student Representative on the Laboratory Life Committee, CMAP, École polytechnique.

2025 Reviewer for ROADEF 2025.

Alpha tester for the *Marmote Markov Decision Process solver*, developed by Emmanuel Hyon and Alain Jean-Marie.

Feb. 2024 Mathematics interviewer for the MSc X–HEC Data Science for Business.

Nov. 2023 Outreach speaker at IHES meeting with Bachelor students from École polytechnique, sharing academic and professional experience in mathematics.

Member of CNRS GDR ROD working groups through MyGDR.

Selected Software and Repositories

GitHub `https://github.com/OrsoF`

service-placement-deployable: PPO-based service placement, routing and bandwidth allocation for edge networks, with feasibility validation and CLI/HTTP interfaces.

service-placement-benchmarks: Python framework for benchmarking latency-constrained service placement in edge networks, comparing reinforcement learning, optimization and heuristics.

mdpforge: Python toolkit for building finite Markov decision processes and benchmarking MDP solvers with reproducible experiments and solution-quality checks.

state_space_disaggregation: Python code for *Progressive State Space Disaggregation for Infinite Horizon Dynamic Programming*.

JMLR2025 Code used to produce experimental results for work on disaggregation techniques for faster MDP solving and state abstraction discovery.

market-forecast Repository on market forecasting using LSTM-based methods.

Technical Skills

Machine Learning: Reinforcement Learning, Deep RL, Hierarchical RL, Approximate Dynamic Programming, State Abstraction, Markov Decision Processes, Planning

Optimization: Combinatorial Optimization, Integer Linear Programming, Network Optimization, Resource Allocation, Dynamic Programming

Programming: Python, NumPy, SciPy, JAX, PyTorch, TensorFlow, Pandas, Gym/Gymnasium, Git, R, Java, C++

Scientific Computing: Sparse matrices, benchmarking, numerical experiments, stochastic simulation, LaTeX, GitHub

Languages

French Native

English Fluent

German Written

Academic Profiles

Website `https://orsof.github.io/`

GitHub `https://github.com/OrsoF`

LinkedIn `https://fr.linkedin.com/in/orso-forghieri`

Scholar `https://scholar.google.com/citations?hl=fr&user=6Vzx6MUAAAAJ`

DBLP `https://dblp.org/pid/376/8414.html`